Small Engines Overhaul · Unit 6 · Level 2 (modified)

Ignition



TODAY'S PLAN

- 🧲 How a magneto works
- ⚡ The spark plug
- 📏 Air gap and plug gap
- 🧪 Testing and troubleshooting

Review · 5 minutes

Quick Review



What does the venturi do?

What does the choke do?

What can ethanol do to a carburetor?

Words to Know

Words to Know



Spark plug

The part that makes a spark across its gap to light the air-fuel mix.



Magneto

The ignition system that uses flywheel magnets and a coil to make high voltage, no battery needed.



Armature (coil)

The coil mounted beside the flywheel that turns the magnets' energy into a high-voltage spark.



Air gap

The small space between the armature and the flywheel magnets, set with a gauge card.

Words to Know

More Words to Know



📏 **Spark plug gap**: The space between the plug's center and side electrodes, set with a gap gauge.

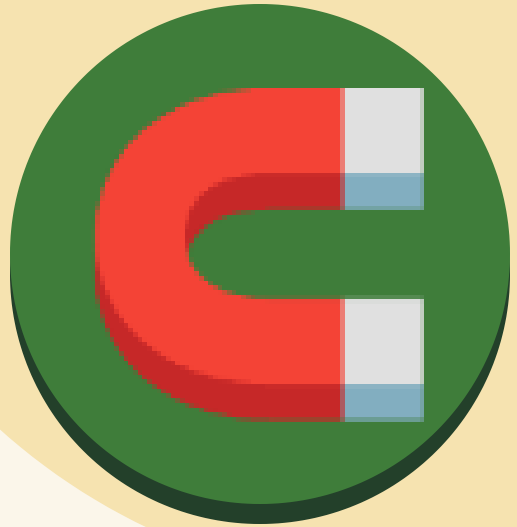
🛞 **Flywheel key**: A soft metal key that holds the flywheel in the right position for spark timing.

🔌 **Kill switch**: A switch or wire that grounds the coil to stop the spark and shut off the engine.

🧪 **Spark tester**: A tool clipped between the wire and ground to safely see if there is a spark.

Learn

The Magneto



Magnets in the flywheel spin past the **armature** (coil).

This makes a small current; the coil boosts it to **thousands of volts**.

The high voltage jumps the plug gap as a spark. No battery needed.

🃏 Check 1 of 2



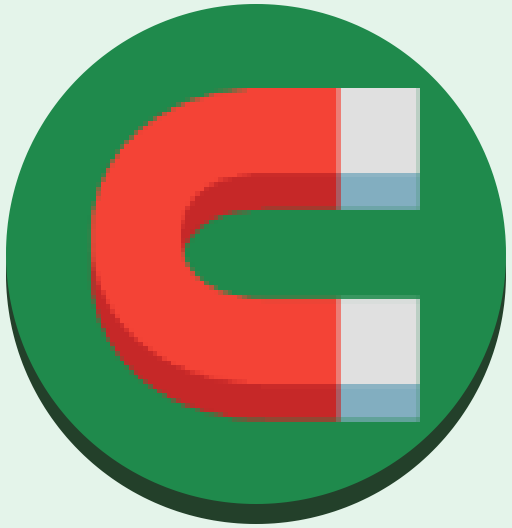
Where are the magneto's magnets?

Hold up 1, 2, or 3:

1️⃣ In the gas tank

2️⃣ In the flywheel

3️⃣ In the muffler



In the flywheel

Flywheel.

Learn

Spark Plug Parts



Terminal

wire connects here



Insulator

white ceramic



Center electrode

spark starts

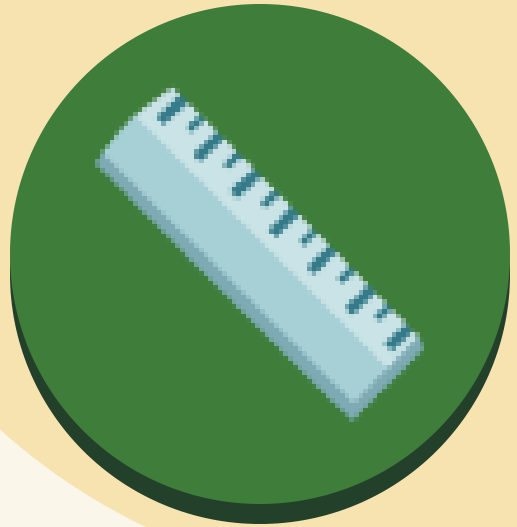


Side electrode

spark jumps to

Learn

Two Gaps to Set



Spark plug gap: set with a gap gauge (often about 0.030 in).

Armature air gap: set with a gauge card (often about 0.010 in).

Always use the **manual's spec**.

Learn

Timing and the Flywheel Key



The **flywheel key** lines up magnets with the piston.

A **sheared key** throws timing off: hard start, kickback.

Common after a mower blade hits a rock.

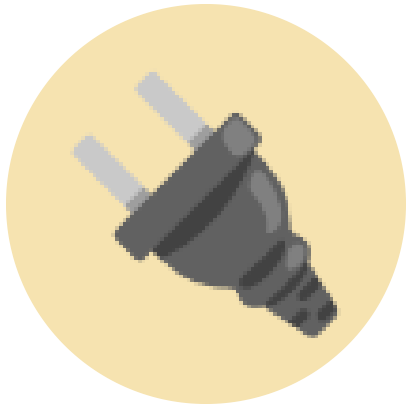
Steps

Spark Test (with adult supervision)



1. Disconnect the plug wire
2. Clip a spark tester between wire and plug/ground
3. Pull the rope
4. Look for a bright blue spark

🃏 Check 2 of 2



No spark. What do you check first?

Hold up 1, 2, or 3:

1️⃣ The muffler

2️⃣ The gas cap

3️⃣ Kill switch wire and plug

✅ Answer · Card 2 of 2



Kill switch wire and plug

Start simple.

Wrap-Up

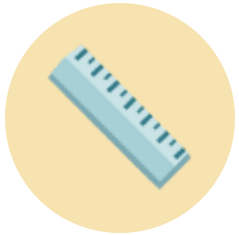
Recap



Magneto
magnets + coil



Plug
spark across gap



Gaps
plug + air gap



Key
timing

Wrap-Up

Exit Ticket



Explain how a magneto makes a spark.

Name the two gaps you set.

What can a sheared flywheel key cause?